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BRANCH : CSM

SECTION : B

COURSE NAME : FULL STACK WEB DEVELOPMENT

CODE : 23CM4121

TITLE OF THE EXPERIMENT : To create a JavaScript-based student profile application using classes, objects, DOM manipulation, dynamic HTML elements, and event handling.

DESCRIPTION: This program creates a Student class with properties such as Name, Roll Number, Department, and CGPA. The user enters the required student details through a form. When the Display Student Profile button is clicked, a Student object is created using the entered values. JavaScript then dynamically creates HTML elements and displays the student profile on the webpage using DOM manipulation. Basic CSS styling is applied to make the generated profile visually clear.

PROGRAM :

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Student Profile</title>

  <!-- Link external CSS file -->

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <div class="container">

    <h1>Student Registration</h1>

    <!-- Student input form -->

    <label for="name">Name:</label>

    <input type="text" id="name" placeholder="Enter student name">
```

```
<label for="rollNo">Roll Number:</label>
<input type="text" id="rollNo" placeholder="Enter roll number">
<label for="department">Department:</label>
<input type="text" id="department" placeholder="Enter department">
<label for="cgpa">CGPA:</label>
<input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">
<!-- Button to generate the profile -->
<button id="displayBtn">Display Student Profile</button>
<!-- Dynamically generated profile will appear here -->
<div id="profileContainer"></div>
</div>
<!-- Link external JavaScript file -->
<script src="script.js"></script>
</body>
</html>
```

Style.css

```
* {
    box-sizing: border-box;
}
body {
    font-family: Arial, sans-serif;
    background-color: #a1939342;
    margin: 0;
    padding: 30px;
}
/* Main container */
.container {
    width: 450px;
    margin: auto;
    padding: 25px;
```

```
    background-color: rgb(98, 123, 124);
    border-radius: 10px;
    box-shadow: 0 4px 12px rgb(54, 104, 73);
}
/* Page heading */
h1 {
    text-align: center;
    color: #070707;
}
/* Form labels */
label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
}
/* Input fields */
input {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #000000;
    border-radius: 5px;
}
/* Display button */
button {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    background-color: #1a1224;
    color: rgb(239, 240, 239);
```

```
border: none;
border-radius: 5px;
font-size: 16px;
cursor: pointer;
}
/* Button hover effect */
button:hover {
    background-color: #306ba9;
}
/* Dynamically generated student profile */
.profile {
    margin-top: 25px;
    padding: 20px;
    background-color: #f2ebf3;
    border-left: 5px solid #3d303d;
    border-radius: 8px;
}
/* Profile heading */
.profile h2 {
    margin-top: 0;
    text-align: center;
    color: #551c45;
}
/* Profile details */
.profile p {
    font-size: 16px;
    line-height: 1.6;
}
Script.js
// Create Student class
```

```

class Student {
    // Constructor initializes student properties
    constructor(name, rollNo, department, cgpa) {
        this.name = name;
        this.rollNo = rollNo;
        this.department = department;
        this.cgpa = cgpa;
    }
}

// Select the button and profile container
const displayButton = document.getElementById("displayBtn");
const profileContainer = document.getElementById("profileContainer");
// Add click event to the button
displayButton.addEventListener("click", function () {
    // Get values entered by the user
    const name = document.getElementById("name").value;
    const rollNo = document.getElementById("rollNo").value;
    const department = document.getElementById("department").value;
    const cgpa = document.getElementById("cgpa").value;
    // Check whether all fields are filled
    if (name === "" || rollNo === "" ||
        department === "" || cgpa === "") {
        alert("Please enter all student details.");
        return;
    }
    // Create an object of Student class
    const student = new Student(
        name,
        rollNo,
        department,

```

```
        cgpa
    );
    // Remove previously displayed profile
    profileContainer.innerHTML = "";
    // Create profile container dynamically
    const profile = document.createElement("div");
    profile.className = "profile";
    // Create heading
    const heading = document.createElement("h2");
    heading.textContent = "Student Profile";
    // Create student detail elements
    const namePara = document.createElement("p");
    namePara.textContent = "Name : " + student.name;
    const rollPara = document.createElement("p");
    rollPara.textContent = "Roll No : " + student.rollNo;
    const departmentPara = document.createElement("p");
    departmentPara.textContent =
        "Department : " + student.department;
    const cgpaPara = document.createElement("p");
    cgpaPara.textContent = "CGPA : " + student.cgpa;
    // Add elements to the profile
    profile.appendChild(heading);
    profile.appendChild(namePara);
    profile.appendChild(rollPara);
    profile.appendChild(departmentPara);
    profile.appendChild(cgpaPara);
    // Display the profile on the webpage
    profileContainer.appendChild(profile);
});
```

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Student Registration

Name:

Roll Number:

Department:

CGPA:

[Display Student Profile](#)

Student Profile

Name : Samhavya
Roll No : A24126552099
Department : CSE(AIML)
CGPA : 8.93